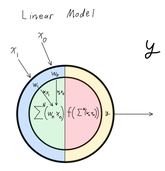
Linear Model



$y = f\left(\sum^{N}(w_n x_n)\right)$

* Similar Properties to Matrices

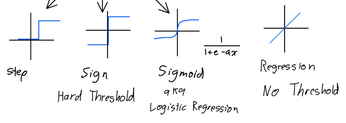
Linear Function Properties

Additivity	Homogeneity
$f(s+x) = f(s)+f(x)$	$f(ax) = af(x)$
$\sum_n^N w_n(s_n+x_n) = \sum_n^N w_n s_n + \sum_n^N w_n x_n$	$\sum_n^N w_n(ax_n) = a\sum_n^N w_n x_n$

Activation Function f

input → f → output

activation



Simple Linear Regression

Model : $y = w_0(x_0)^0 + w_1(x_1)^1$ Dataset : $(x_1, y_1) \ldots (x_n, y_n)$

$= b(1) + mx$

$\boxed{y = b + mx}$

Model

↓

MSE

↓

optimal weights

↓

How do optimal weights differ average weights using standard error (SE)

↓

Account for noise using Residual Standard error (RSE)

↓

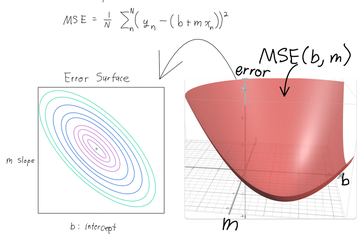
Generate Confidence Interval using SE

Standar error computation

$SE(\hat{m}) = \frac{var(\epsilon)}{\sum(x-\bar{x})^2}$ $\quad SE\left(\begin{matrix}\hat{m}\\ \hat{b}\end{matrix}\right)^2 = var(\epsilon)\left(\begin{matrix}\frac{1}{N} + \frac{\bar{x}^2}{\sum(x-\bar{x})^2}\\ \frac{1}{N}\end{matrix}\right)$

Optimize Model

$MSE = \frac{1}{N}\sum_n^N (y_n - (b+mx_n))^2$



$\frac{dMSE}{dm} = 0$ $\quad \frac{dMSE}{db} = 0$

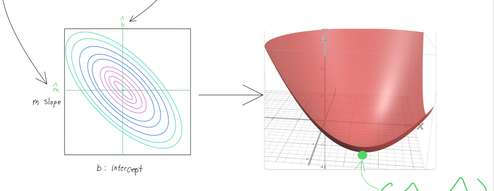
$\bar{x} = E[x] = \frac{1}{N}\sum x$

$\bar{y} = E[y] = \frac{1}{N}\sum y$

$\hat{m} = \frac{N\sum xy - \sum x \sum y}{N\sum x^2 - (\sum x)^2}$ $\quad b = \frac{1}{N}(\sum y - m\sum x)$

$= \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$ $\quad = \frac{1}{N}\sum y - m\left(\frac{1}{N}\sum x\right)$

$\hat{m} = \frac{Covariance(x,y)}{Variance(x)}$ $\quad \hat{b} = \bar{y} - \hat{m}\bar{x}$

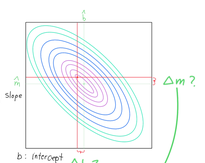


What about Noise?

Effect noise

- optimal model not reachable

$MSE(m,b) = \frac{1}{N}\sum(y-(mx+b)-\xi)^2$



Note $SE^2($

Standard Error

$SE(\hat{m})^2 = \frac{var(\xi)}{\sum(x-\bar{x})^2} = \frac{var(f)}{var(x)}$

$SE(\hat{b})^2 = var(\xi)\left(\frac{1}{N} + \frac{\bar{x}^2}{\sum(x-\bar{x})^2}\right) = var(\xi)\left(\frac{1}{N} + \frac{\bar{x}^2}{var(x)}\right)$

What is var(ξ)?

Residual Sum Error Squared

$var(\xi) \approx (RSE)^2$

$\approx \left(\sqrt{\frac{RSS}{N-M}}\right)^2$

data points

$\approx \frac{RSS}{N-M}$

model parameters

2

(b, m)

Residual Sum Squares

$RSS = N(MSE(\hat{m},\hat{b}))$

$= N\left(\frac{1}{N}\sum^N(y_n - \hat{m}x_n - \hat{b})^2\right)$

$= \sum^N(y_n - \hat{m}x_n - \hat{b})^2$???

$RSS = \sum^N(y_n - mx_n - b - \xi)^2$

RSS example

Best linear model computation

- Consider the dataset d = { p_1:(1,9/2), p_2:(2,5/2), p_3:(3,2) }
- Verify that the best linear model is g = { b: 11/2, m: -5/4 }

$\bar{x} = \frac{1}{3}\sum(1+2+3) = 2$ $\quad \bar{y} = \frac{1}{3}\sum(9/2+5/2+2) = 3$

$\hat{m} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \frac{-1\times 3/2 + 0\times(-1/2) + 1\times(-1)}{(-1)^2+0^2+(-1)^2} = -\frac{5}{4}$

$\hat{b} = \bar{y} - \hat{m}\bar{x} = 3 - (-5/4)\times 2 = 11/2$

$\hat{y} = \frac{11}{2} - \frac{5}{4}x$

$RSS = \sum(y - \hat{m}x - \hat{b})^2$

$= \left(\frac{9}{2} - \left(-\frac{5}{4}(1) + \frac{11}{2}\right)\right)^2 + \left(\frac{5}{2} - \left(-\frac{5}{4}(2) + \frac{11}{2}\right)\right)^2 + \left(2 - \left(-\frac{5}{4}(3) + \frac{11}{2}\right)\right)^2$

$= \left(\frac{1}{4}\right)^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2$

$RSS = \frac{1}{16} + \frac{1}{4} + \frac{1}{16}$

$= \frac{1}{8} + \frac{2}{8}$

$RSS = \frac{3}{8}$

$var(\xi)$ $\quad \frac{3/8}{1} = \frac{3}{8}$

$SE(\hat{b})^2 = var(\xi)\left(\frac{1}{N} + \frac{\bar{x}^2}{var(x)}\right)$

$= \frac{3}{8}\left(\frac{1}{3} + \frac{(2)^2}{2}\right)$

$SE(\hat{b})^2 = \frac{3}{8}\left(\frac{1}{3} + 2\right)$

$SE(\hat{b}) = \sqrt{\frac{3}{8}\left(\frac{1}{3}+2\right)}$

Confidence interval

B% confidence interval : $\frac{B}{100}$ probability the parameter's unknown value in a given range

Parameters : $\hat{w}$

Confidence : 95%

Where does 2 come from ???

95% Confidence Interval : $[\hat{w} - 2SE(\hat{w}), \hat{w} + 2SE(\hat{w})]$

Ex

Parameters : $\{w_0 \ldots w_N\}$ 95% confidence

w_0 : $CI = [\hat{w}_0 - 2SE(\hat{w}_0), \hat{w}_0 + 2SE(\hat{w}_0)]$

$\vdots$

w_N : $CI = [\hat{w}_N - 2SE(\hat{w}_N), \hat{w}_N + 2SE(\hat{w}_N)]$

How can we verify the calculated Intervals are trustworthy?

Statistical Significance = Trustworthy???

t-test and p-values

Disproving an opposing Hypothesis

Null Hypothesis

H_0 : There is relationship between x and y aka aka

$Covariance(x,y) = 0$

$\hat{m} = \frac{cov(x,y)}{var(x)} = 0$,

$w = 0$

For the linear Regression $y = w_0 + w_1 x$

where $w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

SO

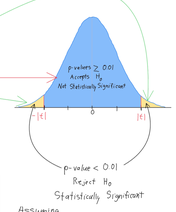
$y = 0 + 0x$ should y for all of x ???

t-statistic (2 tail distribution)

$t = \frac{\hat{w} - 0}{SE(\hat{w})}$

$p\text{-value} = P(|T| \geq t \mid H_0)$

We'll assume the Null Hypothesis has Normal Probability Distribution



Assuming

$\hat{m}$: $t = -2.8828$

$\hat{b}$: $t = 5.8798$

degree of freedom = # data points − # parameters

= 3 − 2

= 1

Assuming,

$t = 637.52$ for $\hat{m}$

$t = 640.39$ for $\hat{b}$

Statistically Significant

one-tailed α	0.10	0.05	0.025	0.01	0.005	0.0005
two-tailed α	0.20	0.10	0.05	0.02	0.01	0.001
df						
1	3.078	6.314	12.71	31.82	63.66	636.62
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.924
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.869
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.408
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073

ed α	0.10	0.05	0.025	0.01	0.005
ed α	0.20	0.10	0.05	0.02	0.01
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787

T-Distribution Table of Critical Values

With trusted intervals

How do we assess the model's overall accuracy?

Evaluating Overall Model Accuracy

Residual Standard Error

$RSE = \sqrt{\frac{RSS}{N-M}}$

- Error between model & data
- the smaller the better

Good — Small ; Bad — large

R squared Statistic

$R^2 = 1 - \frac{RSS}{TSS}$

$= 1 - \frac{\sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$ How does this show variability of the data

$= 1 - \frac{\sum(y-\hat{m}x-\hat{b})^2}{\sum(y-\bar{y})^2}$ How much model can explain the relation x and y

$R^2 = 1 - \frac{sum\ squared\ error}{N \cdot var(y)}$???

- Goal : How much a model can explain relation between X and y
- "Proportion of output variability explained by the input"
- Relationship between label and prediction over

F-statistic

$TSS = \sum(y-\bar{y})^2 = \sum\left(y - \frac{1}{N}\sum y\right)^2$

$F = \frac{(TSS - RSS)(N-M)}{(RSS)(M-1)}$

- Tests H_0 where all parameters are 0

no relation — 1 ; Good relation — large

Adjusted R^2

$1 - \frac{(1-R^2)(N-1)}{(N-M)}$

Not Explainable — 0 ; Explainable — 1